Nirmol — Demo Script

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Opening note: a mobile app, shown as a web app

Nirmol is designed as a **mobile app**: people check the air before leaving home and report smoke or dust from the street, so it belongs on their phones. Because of the hackathon time limit we built it as a **mobile-first web app** instead. It runs in any phone browser with the same screens, and it can later be packaged as an installable app (PWA) or rebuilt natively with the same backend.

Say this in the first 20 seconds of the demo:

> "Nirmol is meant to be a mobile app. With the time we had, we built it as a mobile-first web app. Everything you'll see works on a phone browser today, and the same backend will power the native app."

Run the demo with the browser in phone view (or on a real phone) so judges see it the way users will.

Before the demo (30 minutes before)

- [ ] Open the live site on a phone, or a laptop browser in phone view (F12 → device toolbar → iPhone/Pixel).
- [ ] Language set to **English** (Bangla shown later as a switch).
- [ ] Sign in once with the demo account, or tap Continue with Google, so login is fast. The menu: 9 tabs across the top, How to use and My Nirmol at the top right.
- [ ] A second account is a **moderator** and is signed in on another browser or device.
- [ ] At least 3 sample reports are already **approved** so the Community Map is not empty.
- [ ] One real pollution photo saved on the phone (rubbish burning, construction dust or black smoke; no faces or number plates).
- [ ] GitHub → Actions → **Hourly forecast** shows a green run in the last hour, so the Nirmol model is live at "Dhaka – Baridhara".
- [ ] Slide with the model results ready: **36% more accurate than CAMS**, **CAMS shows 81 vs real 170 µg/m³**.
- [ ] Backup: the 3–4 minute screen recording of the full demo, saved offline.
- [ ] Phone on Wi-Fi **and** mobile data available (the venue Wi-Fi may drop).

Who presents what

The demo runs about **6 minutes** in three parts; each presenter owns the part closest to what they built. Fill in names below.

Part	Presenter	Time	Covers
1. Problem and live air	A (ML / data) — name:	0:00–2:00	Mobile-app note, the problem, dashboard, Nirmol forecast and "why"
2. Personal decisions	B (frontend) — name:	2:00–4:00	Exposure, best time to leave, alerts, AI advisor in Bangla
3. Community action	C (backend + AI) — name:	4:00–5:30	Report with photo + AI, moderation, Community Map
Wrap-up	A	5:30–6:00	What's special, what's next, thank you

Whoever is not speaking drives the phone, so the speaker can face the judges.

The script

Each step says what to **do** on screen and what to **say**. Speak to the judges, not the screen.

Part 1 — Problem and live air (Presenter A, 2 min)

1. **Do:** show the dashboard, Dhaka selected. **Say:** "Nirmol is meant to be a mobile app; in the time we had, we built it as a mobile-first web app. Every winter, Dhaka's air is among the worst in the world. People know the air is bad, but they don't know *when* to go out, *how long* is OK, or *what to do* about the smoke they see. Nirmol answers those three questions."
2. **Do:** point at the big PM2.5 number, the coloured level and the advice line. **Say:** "This is the air right now: PM2.5, an AQI estimate and one clear action, in plain words. Every number says where it comes from and how fresh it is; we never show a model estimate as a measurement."
3. **Do:** read the yellow warning. **Say:** "Here's what we found: global models like CAMS show Dhaka's winter air at about **half** its real level. At the US Embassy monitor, real winter PM2.5 averaged **170**, while CAMS showed **81**."
4. **Do:** switch the place to **Dhaka – Baridhara (Nirmol model)**; show the forecast chart with the shaded range. **Say:** "So we trained our own model. It corrects CAMS using two and a half years of embassy data. On a winter it had never seen, it is **36% more**

accurate than CAMS. The shaded band is where the real value falls about 80% of the time."

5. **Do:** pick an hour in "Why this forecast?". **Say:** "And it explains itself: these are the three things that pushed this hour up or down, like cold temperature trapping smog or low wind."

Part 2 — Personal decisions (Presenter B, 2 min)

6. **Do:** open **Exposure**, choose Walking, 2 hours, tick "Breathing problems". **Say:** "The same air affects people differently. Nirmol combines pollution, time outside and activity into one exposure score, and is more careful for children, older people and people with asthma."
7. **Do:** open **Travel planner**, set a 6-hour window and a 1-hour trip. **Say:** "Instead of just 'the air is bad', Nirmol tells you the best time to leave. Here, leaving at this time means the least pollution among your options. And with Compare plans, you can put two plans side by side, like walking now versus a rickshaw later."
8. **Do:** open **Alerts**, set a limit of 75. **Say:** "Set your own limit, and Nirmol warns you when the air now or later goes over it."
9. **Do:** open **AI advisor**, tap the language switch to **বাংলা**, ask: *kalke shokale hata ki bhalo hobe?* **Say:** "People can ask in Bangla, English or even Banglish. The answer uses real air data and checked advice; the AI never invents numbers or gives medical diagnoses."
10. **Do:** ask *buke betha hocche*. **Say:** "And for emergencies like chest pain, it doesn't wait for AI: it immediately says to call 999."

Part 3 — Community action (Presenter C, 1.5 min)

11. **Do:** open **Report pollution**, add the photo, tap **Get AI suggestion**. **Say:** "When someone sees rubbish burning or construction dust, they can report it. AI suggests the type from the photo, but the person confirms it. Hidden photo data like GPS tags is removed, users are asked not to photograph faces or number plates, and the exact spot stays private."
12. **Do:** pick the location, tick the consent boxes, **Send report**. **Say:** "Reports are private until a moderator checks them, so the map can't be spammed or used to accuse people."
13. **Do:** on the moderator device, open **Moderation**, approve the report. **Say:** "A moderator reviews and approves it, with a full record of every decision."
14. **Do:** open **Community Map**. **Say:** "Approved reports appear here as rough 5 km areas, never exact addresses. Over time this shows where pollution sources cluster, which is useful for city authorities and NGOs."

Wrap-up (Presenter A, 30 s)

15. **Say:** "Nirmol turns air data into decisions: when to go out, how long, and what to do about pollution you see. It's honest about uncertainty, works in Bangla, and protects privacy. Next, we want to make it a full mobile app with SMS alerts and neighbourhood sensors. Thank you."

Features at a glance

Fourteen features are built and working in the web app; route comparison and SMS/Telegram alerts are not built yet.

Feature	Page	What it does
Live air quality	Dashboard	PM2.5, AQI estimate, weather and plain advice for about 70 cities and Dhaka/Chattogram neighbourhoods or your GPS location, each value labelled with its source and age
AI forecast	Dashboard	24-hour forecast; at Baridhara, Dhaka it uses Nirmol's own trained model (36% more accurate than CAMS), elsewhere CAMS with a warning
Explainable forecast	Dashboard	"Why this forecast?": the 3 factors behind each hour (e.g. temperature, wind)
Personal exposure	Exposure	Exposure score from PM2.5 × hours × activity, with extra care for children, older people and health conditions
Best time to leave	Travel planner	Finds the departure time with the least pollution within your window
Compare plans	Compare plans	Plan A vs Plan B side by side (leave now vs later, walk vs rickshaw) to see which means less pollution
Alerts	Alerts	Warns in the app when the air now or later passes your own limit
Exposure diary	Diary	Private log of time outside and symptoms, kept only in the browser, exportable
Pollution reporting	Report pollution	Photo report with AI-suggested type (Gemini), private exact location, hidden photo data removed

Feature	Page	What it does
Moderation	Moderation	Moderators approve or reject reports, with a full decision history
Community Map	Community	Approved reports on a map as rough 5 km areas, filterable by type and date
Bangla AI advisor	AI advisor	Ask in Bangla, English or Banglish; answers from real air data and checked advice, 999 for emergencies, 20 questions a day
Accounts	Sign in	Email or Google sign-in; save places and preferences
Bilingual, light/dark	Every page	English default, everyday Bangla, phone-first layout, light and dark themes

What makes Nirmol special

Other air apps show a number; Nirmol turns it into a decision and an action. Stress these five points:

1. **Our own model fixes a real gap.** Global models show Dhaka's winter air at about half its real level (CAMS 81 vs measured 170 µg/m³). Nirmol's model, trained on US Embassy data, is 36% more accurate than CAMS on an unseen winter, with a calibrated range (81% coverage for an 80% band).
2. **From information to decisions.** Best time to leave, personal exposure, plan comparisons and alerts answer "what should I do?", not just "how bad is it?".
3. **Citizens become sensors.** Photo reports with AI suggestions, human moderation and a privacy-safe community map turn what people see into data for authorities and NGOs.
4. **Built for Bangladesh.** Everyday Bangla, Banglish questions, Bangladeshi places, 999 for emergencies and brick kilns as a pollution type.
5. **Honest and private by design.** Every number shows its source and age, the AI never invents numbers or gives diagnoses, exact report locations and health notes stay private, and the explanation shows *why* the model predicts what it does.

What can be extended next

The first step is the mobile app itself; everything else builds on the same backend.

Next step	What it adds
Native mobile app (or installable PWA)	Home-screen app, push notifications, offline last-known air, camera reporting in one tap
SMS and Telegram alerts	Warnings for people without smartphones or data, in Bangla
Neighbourhood sensors	Low-cost PM2.5 sensors (e.g. PMS5003 + ESP32), calibrated against reference monitors, for street-level accuracy and live measurements
Model for every city	Train and test the correction model for Chattogram, Sylhet, Rajshahi and others as monitoring data allows
Pollution-aware route comparison	Compare routes once there is enough street-level data to tell routes apart
School and workplace dashboards	Outdoor-activity advice for schools, factories and offices
Authority dashboard	Hotspot clusters and exported reports for the Department of Environment, city corporations and NGOs
Report follow-up	Track a report from approved to forwarded to resolved
Symptom patterns	Optional, consented health logging to study links between exposure and symptoms (with doctors)

Judge questions and backup plan

If a judge asks…	Answer
"Is your forecast better than just repeating the current value?"	Better than repeating yesterday's value (error 51 vs 61 µg/m³). Repeating the last hour is better, but impossible live: Bangladesh has had no live reference monitor online since the US Embassy feed stopped in March 2025.

If a judge asks…	Answer
"Isn't this just CAMS?"	No. CAMS shows about half the real winter level in Dhaka. Our model learned the correction from 2.5 years of embassy data and cuts the error by 36%.
"Why only Baridhara for your model?"	That's the only site with reliable measurements to test against. Showing it elsewhere would be a claim we can't prove, so other places use CAMS with a clear warning.
"Why not a mobile app?"	Time limit. We built mobile-first so the same screens and backend move straight into an app.
"Can people misuse reports?"	Reports stay private until a moderator approves them; the public map only shows rough 5 km areas, and there are daily limits.
"Does the AI make things up?"	It can only pick from checked advice and the real numbers we give it. Emergencies like chest pain skip the AI and say call 999.
"What does it cost to run?"	Nothing so far: everything runs on free tiers (Vercel, Supabase, GitHub Actions, Open-Meteo, Gemini free tier).

If something fails during the demo:

- *Air data won't load:* tap Try again once; if it still fails, say "the venue internet dropped" and switch to the recorded video.
- *Advisor says busy:* Gemini's free tier is overloaded; show the emergency answer (no AI needed) and move on.
- *Login fails:* skip it; dashboard, exposure, travel, alerts and the map all work without an account.
- *Map is empty:* show the moderation screen instead and explain the approval step.